

Diffused Laplace AR(1):

exact $K = 1$ benchmark, nonlinear score, and state-dependent effective precision

March 2026

Hypothesis -> derivation -> signal -> conclusion.

1. **Hypothesis.** Relative to the Gaussian benchmark, the first robust non-Gaussian signal should be that $-\nabla^2 \log p_t(x)$ is no longer constant.
2. **Derivation.** For $K = 1$, the Laplace model admits an exact one-dimensional reduction after OU noising. From it one gets closed formulas for the density, the score, and the curvature matrix $\mathcal{H}_t(x) := -\nabla_x^2 \log p_t(x)$.
3. **Signal.** The score is nonlinear and $\mathcal{H}_t(x)$ varies with x .
4. **Conclusion.** Finite-time smoothing does *not* Gaussianize the Laplace model. The old draft's stronger $K = 2$ “closed-form” claim is not retained.

1 Model and the benchmark question

We study the two-frame Laplace AR(1) model

$$A_0 \sim \mathcal{N}(\mu_0, \sigma_0^2), \quad A_1 = \alpha A_0 + \varepsilon, \quad \varepsilon \sim \text{Laplace}(0, b), \quad |\alpha| < 1, \quad (1)$$

followed by independent OU noising

$$X_k = e^{-t} A_k + \sqrt{\Delta_t} Z_k, \quad \Delta_t = 1 - e^{-2t}, \quad Z_0, Z_1 \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1). \quad (2)$$

The matched Gaussian reference replaces ε by $\varepsilon_G \sim \mathcal{N}(0, 2b^2)$, so the clean first two moments match exactly. In that Gaussian model,

$$\nabla \log p_t^G(x) = -Q_t(x - \mu_t), \quad -\nabla^2 \log p_t^G(x) \equiv Q_t, \quad (3)$$

where Q_t is the noisy precision matrix. The whole Gaussian geometry is constant in x .

The question is therefore simple:

After finite OU smoothing, does the Laplace model still retain enough structure for $-\nabla^2 \log p_t(x)$ to depend on x ?

2 Minimal Gaussian baseline

For the matched Gaussian model the clean mean and covariance are

$$m_A = \begin{pmatrix} \mu_0 \\ \alpha \mu_0 \end{pmatrix}, \quad \Sigma_A = \begin{pmatrix} \sigma_0^2 & \alpha \sigma_0^2 \\ \alpha \sigma_0^2 & \alpha^2 \sigma_0^2 + 2b^2 \end{pmatrix}. \quad (4)$$

The cusp is smoothed immediately, but finite-time non-Gaussianity remains

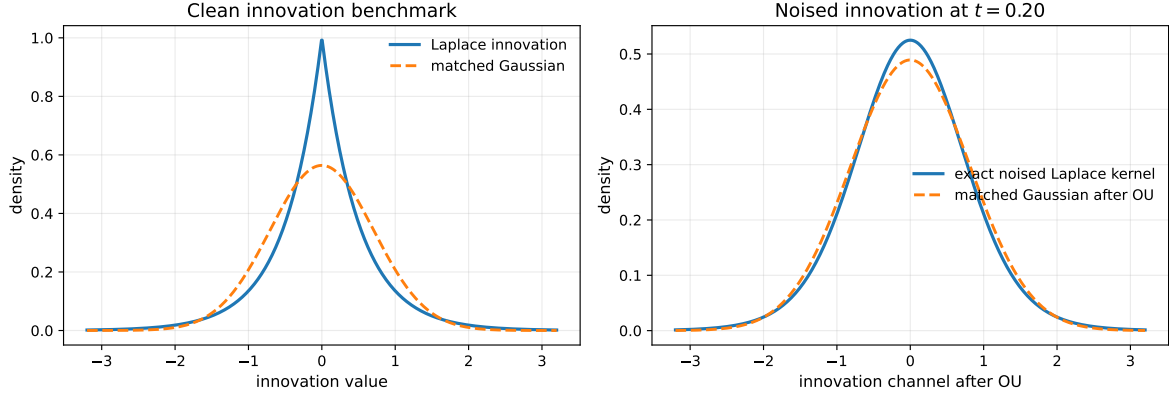


Figure 1: **Innovation-level benchmark.** Left: the clean Laplace innovation and the matched Gaussian innovation with the same variance $2b^2$. Right: the exact noised innovation density $h(y; \beta_t, \Delta_t)$ at $t = 0.20$. OU noise smooths the cusp immediately, but the finite-time law is still not Gaussian. Parameters: $\alpha = 0.8$, $b = 0.5$, $\mu_0 = 0$, and $\sigma_0^2 = 2b^2/(1 - \alpha^2)$.

After OU noising,

$$\mu_t = e^{-t}m_A, \quad \Sigma_t = e^{-2t}\Sigma_A + \Delta_t I_2, \quad Q_t = \Sigma_t^{-1}. \quad (5)$$

Hence the benchmark curvature is exactly constant:

$$\mathcal{H}_t^G(x) \equiv Q_t.$$

This is the object that the Laplace model has to beat.

3 Exact $K = 1$ Laplace reduction

Step 1: the noised innovation kernel

Let $\beta_t = e^{-t}b$. The diffused innovation $e^{-t}\varepsilon + \sqrt{\Delta_t}Z$ has density

$$h(y; \beta_t, \Delta_t) = \frac{e^{\Delta_t/(2\beta_t^2)}}{2\beta_t} \left[e^{-y/\beta_t} \Phi\left(\frac{y}{\sqrt{\Delta_t}} - \frac{\sqrt{\Delta_t}}{\beta_t}\right) + e^{y/\beta_t} \Phi\left(-\frac{y}{\sqrt{\Delta_t}} - \frac{\sqrt{\Delta_t}}{\beta_t}\right) \right]. \quad (6)$$

This is the exact Laplace-Gaussian convolution. It is smooth for every $t > 0$, but it is not Gaussian at finite time.

Step 2: condition on the first coordinate

The first noisy coordinate is Gaussian:

$$X_0 \sim \mathcal{N}(e^{-t}\mu_0, v_x), \quad v_x = e^{-2t}\sigma_0^2 + \Delta_t. \quad (7)$$

Moreover,

$$A_0 \mid X_0 = x_0 \sim \mathcal{N}(m_t(x_0), v_t), \quad v_t^{-1} = \sigma_0^{-2} + e^{-2t}\Delta_t^{-1}, \quad m_t(x_0) = v_t \left(\frac{\mu_0}{\sigma_0^2} + \frac{e^{-t}x_0}{\Delta_t} \right). \quad (8)$$

If $c_t = e^{-t}\alpha$, then the exact noisy density is

$$p_t(x_0, x_1) = \phi_{v_x}(x_0 - e^{-t}\mu_0) \mathbb{E}[h(x_1 - c_t A_0; \beta_t, \Delta_t) \mid X_0 = x_0], \quad (9)$$

where $\phi_v(u) = \frac{1}{\sqrt{2\pi v}} e^{-u^2/(2v)}$.

So the whole $K = 1$ problem is reduced to a *one-dimensional Gaussian expectation*. No Monte Carlo approximation is needed.

4 Exact score and exact effective precision

For $r = 0, 1, 2$, define

$$I_r(x_0, x_1; t) := \mathbb{E} \left[h^{(r)}(x_1 - c_t A_0; \beta_t, \Delta_t) \mid X_0 = x_0 \right]. \quad (10)$$

Then $p_t(x_0, x_1) = \phi_{v_x}(x_0 - e^{-t}\mu_0)I_0(x_0, x_1; t)$, and the exact score is

$$\partial_{x_1} \log p_t(x_0, x_1) = \frac{I_1}{I_0}, \quad (11)$$

$$\partial_{x_0} \log p_t(x_0, x_1) = -\frac{x_0 - e^{-t}\mu_0}{v_x} - c_t m'_t(x_0) \frac{I_1}{I_0}, \quad m'_t(x_0) = \frac{v_t e^{-t}}{\Delta_t}. \quad (12)$$

Now define the effective precision

$$\mathcal{H}_t(x) := -\nabla_x^2 \log p_t(x). \quad (13)$$

A direct differentiation gives

$$(\mathcal{H}_t(x))_{11} = \frac{I_1^2 - I_0 I_2}{I_0^2}, \quad (14)$$

$$(\mathcal{H}_t(x))_{01} = -c_t m'_t(x_0) (\mathcal{H}_t(x))_{11}, \quad (15)$$

$$(\mathcal{H}_t(x))_{00} = \frac{1}{v_x} + c_t^2 (m'_t(x_0))^2 (\mathcal{H}_t(x))_{11}. \quad (16)$$

In the Gaussian case these are constants. In the Laplace case they are functions of x .

Why this is a genuinely non-Gaussian signal. If $\mathcal{H}_t(x)$ were constant on all of $\mathbb{R}^2$, then $\nabla \log p_t(x)$ would be affine, hence $\log p_t(x)$ would be a quadratic polynomial and p_t would be Gaussian. But the exact characteristic function is

$$\Phi_t(\xi_0, \xi_1) = \exp \left(i e^{-t} \mu_0 (\xi_0 + \alpha \xi_1) - \frac{1}{2} \Delta_t (\xi_0^2 + \xi_1^2) - \frac{1}{2} e^{-2t} \sigma_0^2 (\xi_0 + \alpha \xi_1)^2 \right) \frac{1}{1 + b^2 e^{-2t} \xi_1^2}, \quad (17)$$

and the rational factor $(1 + b^2 e^{-2t} \xi_1^2)^{-1}$ prevents Gaussianity at every finite t . Therefore $\mathcal{H}_t(x)$ cannot be constant.

5 Conclusion on the hypotheses

1. **Finite-time smoothing does not erase non-Gaussianity.** Equation (17) shows that the diffused Laplace law is not Gaussian for any finite t .
2. **The first robust signal is x -dependent curvature.** Equations (14)–(16) give exact formulas for $\mathcal{H}_t(x)$, and Figures 3–4 show the variation directly.
3. **The old $K = 2$ closed-form claim is not retained.** The corrected statement is weaker: $K = 2$ has an exact nested integral representation, but the previous draft did not establish an elementary closed form comparable to (9).

The presentation message is therefore short and clean:

Gaussian benchmark $\implies$ constant Q_t , Laplace benchmark $\implies$ exact $\mathcal{H}_t(x)$ depending on x .

That is the main signal to present.

The exact Laplace score is nonlinear; the Gaussian score is affine

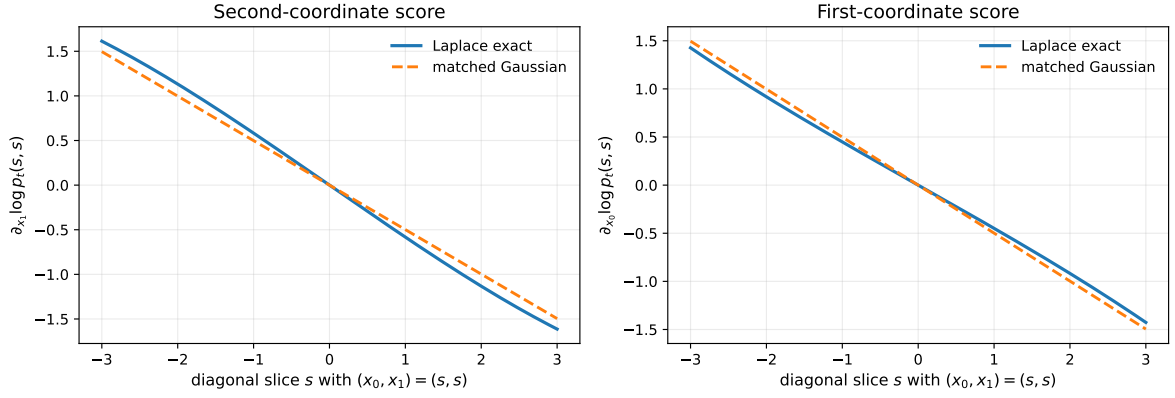


Figure 2: **Score slices.** Along the diagonal slice $(x_0, x_1) = (s, s)$, the exact Laplace score is nonlinear, while the matched Gaussian score is affine. This is the first visible signal that the Gaussian benchmark has been left behind.

For Laplace innovations the effective precision depends on position

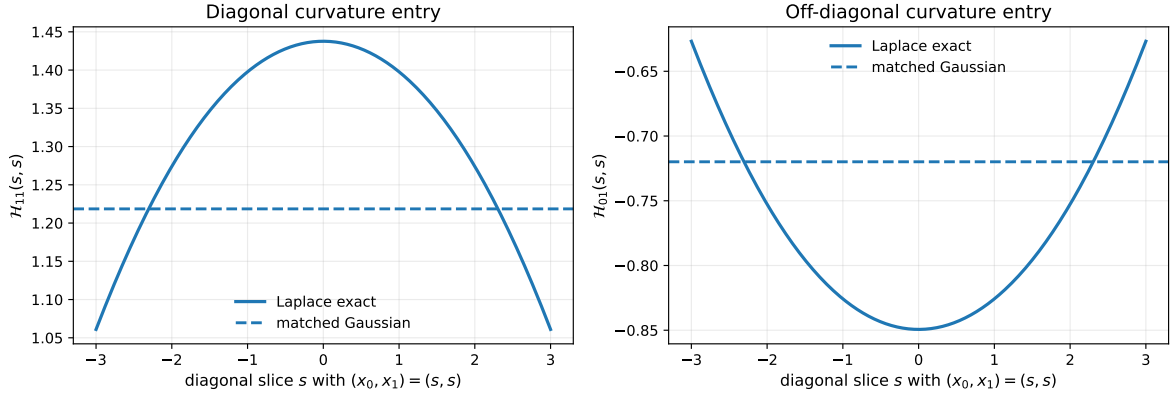


Figure 3: **Curvature on the diagonal slice.** Along $(x_0, x_1) = (s, s)$, both the diagonal and the off-diagonal entries of $\mathcal{H}_t(s, s)$ vary with s . In the matched Gaussian model the corresponding entries are constant.

The Gaussian constant precision is replaced by a full x -dependent curvature field

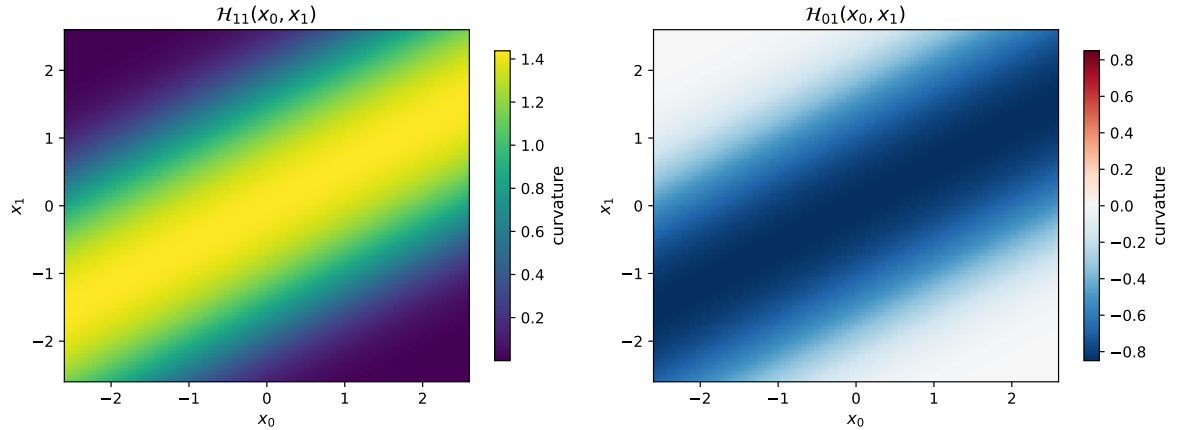


Figure 4: **Two-dimensional curvature field.** Heatmaps of the exact entries $(\mathcal{H}_t(x_0, x_1))_{11}$ and $(\mathcal{H}_t(x_0, x_1))_{01}$. The Gaussian notion of one fixed precision matrix is replaced by an x -dependent field.